



SwapSkill

Skill Content Summary

Category: Database Management

Skill: PostgreSQL

Topics Breakdown

- Introduction to PostgreSQL
 - ! Features
 - ! Advantages
 - ! Use Cases
- PostgreSQL Architecture
 - ! Client Server Model
 - ! Processes
 - ! Memory Structure
- Installation and Setup
 - ! Installing PostgreSQL
 - ! pgAdmin
 - ! Configuration
- Database Basics
 - ! Databases
 - ! Schemas
 - ! Tables
- Data Types in PostgreSQL
 - ! Numeric Types
 - ! Character Types
 - ! Date and Time Types
- Table Management
 - ! Create Table
 - ! Alter Table
 - ! Drop Table
- Constraints
 - ! NOT NULL
 - ! UNIQUE
 - ! PRIMARY KEY
 - ! FOREIGN KEY
 - ! CHECK
- Keys
 - ! Primary Key
 - ! Foreign Key
 - ! Composite Key
- CRUD Operations
 - ! INSERT
 - ! SELECT

- ! ' UPDATE

- ! ' DELETE

- Querying Data

- ! ' WHERE Clause

- ! ' ORDER BY

- ! ' LIMIT

- ! ' OFFSET

- Operators

- ! ' Arithmetic Operators

- ! ' Comparison Operators

- ! ' Logical Operators

- Functions

- ! ' Built

- Joins

- ! ' Inner Join

- ! ' Left Join

- ! ' Right Join

- ! ' Full Join

- Subqueries

- ! ' Single Row Subquery

- ! ' Multiple Row Subquery

- Indexes

- ! ' Types of Indexes

- ! ' Creating Indexes

- ! ' Performance Impact

- Views

- ! ' Creating Views

- ! ' Updating Views

- Sequences

- ! ' Auto Increment

- ! ' Serial

- ! ' Identity Columns

- Transactions

- ! ' BEGIN

- ! ' COMMIT

- ! ' ROLLBACK

- ! ' Savepoints

- ACID Properties

- ! ' Atomicity

- ! ' Consistency

- ! ' Isolation

- ! ' Durability

- Normalization

- ! ' First Normal Form

- ! ' Second Normal Form

- ! ' Third Normal Form

- Stored Procedures

- ! ' Functions

- ! ' Procedures

- ! ' Parameters

- Triggers

- ! ' Before Trigger
- ! ' After Trigger
- ! ' Trigger Functions
- Security
 - ! ' Roles
 - ! ' Privileges
 - ! ' User Management
- Backup and Restore
 - ! ' pg_dump
 - ! ' pg_restore
- Performance Optimization
 - ! ' Query Optimization
 - ! ' Explain A